Paragon TRITIUM: The H3 Enterprise Spine

A Blueprint for Global Expansion — Every Store, Every Consumer, Every Rupiah on One Hexagonal Grid

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Abstract. Where should the next store open? Which consumers does the network already reach? Where is revenue leaking to unserved demand? Every one of these decisions is keyed to a map — and in Indonesia that map is usually administrative geography: polygons that differ in area by three orders of magnitude, cut catchments at political borders, and are periodically redrawn (*pemekaran*), breaking every year-over-year comparison. Paragon TRITIUM replaces them with an *H3 Enterprise Spine*: every store, resident, consumer, and rupiah of revenue indexed to the same fixed hexagonal grid at ingestion, turning spatial analysis into ordinary SQL joins on the data warehouse the company already runs. On a reproducible Jakarta–Tangerang corridor, the spine isolates unserved *whitespace* worth ∼IDR 8.4 bn per month and reaches 50% of the population with 25% of the land area, where district targeting needs 38% — leaner media spend, sharper expansion shortlists, measurable delivery SLAs. And because the grid covers the planet, the same schema, SQL, and playbook deploy unchanged in every new market: a working blueprint for global expansion, delivered in three phases with no new infrastructure.

Keywords: H3 spatial indexing, hexagonal grids, geospatial analytics, market penetration, whitespace analysis, retail gravity, MAUP, Indonesia.

1 Introduction

Every strategic question a retail and distribution business asks about territory reduces to joining four layers of data at a common spatial unit: where the *stores* are, where the *population* lives, how *consumers* in each place actually behave, and where *revenue* is — and is not — being captured. The choice of that spatial unit is usually treated as an afterthought, inherited from whatever polygons the data arrived in. In Indonesia those polygons are almost always administrative: 38 provinces, 500-plus *kota/kabupaten*, some 7,000 *kecamatan*, and over 80,000 *desa/kelurahan*.

This inheritance is not neutral. Administrative units vary in size by three orders of magnitude, so any per-unit statistic — revenue per capita, outlets per district, penetration rate — is confounded by the unit's arbitrary extent, the classical *modifiable areal unit problem* (MAUP) [5]. Their borders cut across organically connected commercial catchments: a store on the western edge of Jakarta draws much of its demand from Tangerang, in a different province. And the units themselves are unstable — *pemekaran*, the recurrent splitting of provinces, regencies, and villages, silently breaks every historical year-over-year comparison keyed to them.

This paper argues for, and specifies, a different foundation: a shared *H3 Spine*. H3 [1] is an open-source hierarchical hexagonal discrete global grid that assigns every coordinate on Earth a short alphanumeric index at each of 16 resolutions. Indexing every fact table — store registry, population raster, consumer records, transaction revenue — to the same H3 resolution at ingestion converts geospatial analysis from expensive polygon geometry into standard SQL string joins, executable natively in modern cloud warehouses and renderable directly in mainstream GIS and BI tooling [3, 10].

Our contributions are fourfold:

1. **A failure analysis of administrative geography** for market analytics in the Indonesian context — MAUP distortion, cross-border catchment blindness, and *pemekaran* instability (Section 3).
2. **The H3 Spine architecture**: a governed, warehouse-native pipeline that standardizes all four data layers onto one hexagonal dimension table (Section 4).
3. **A formal metrics layer** on the spine: composite demand mass, distance-decay revenue capture, a four-class penetration taxonomy isolating whitespace, k-ring cannibalization, and area–reach gain curves (Section 5).
4. **An end-to-end case study** on a deter-

ministic synthetic corridor modeled on western Jakarta–Tangerang, quantifying targeting gains and whitespace opportunity (Sections 6 and 7).

The framework is deliberately engine-agnostic downstream: the same spine that answers the strategic questions here feeds the context-aware routing layer of our companion work on fleet optimization [12].

2 Background: The H3 Discrete Global Grid

2.1 Mechanics

H3 tessellates the globe into hexagonal cells at 16 nested resolutions, from resolution 0 ($\sim$4.25 $\times$ 10^6 km^2 per cell) down to resolution 15 ($\sim$0.9 m^2) [2]. Three properties make it the natural unit for market analytics.

Uniform, near-isotropic adjacency. Unlike squares — whose diagonal neighbors are $\sqrt{2}$ farther than their edge neighbors — a hexagon's six neighbors all lie at the same center-to-center distance, minimizing directional bias in catchment and proximity computations [4]. At a fixed resolution cells are near-uniform in area (resolution 8, the workhorse of this paper, averages 0.737 km^2), so per-cell counts are directly comparable — the property administrative polygons conspicuously lack.

Hierarchical indexing. Each parent cell subdivides into approximately seven children one resolution finer (aperture 7, Figure 1). Because the hierarchy is encoded in the index itself, coarsening an analysis from neighborhood to regional scale is an index-truncation operation — no re-projection, no polygon dissolve. Resolution can therefore be treated as a *query parameter* rather than a data-modeling commitment: the same resolution-8 spine rolls up to resolution 6 for provincial dashboards or drills to resolution 10 ($\sim$15,000 m^2) for site selection.

Index-as-key computation. An H3 index is a short string. Point data (a POS transaction, a customer address, a competitor location) is assigned its cell once at ingestion; thereafter every spatial operation — joining revenue to population, counting stores near a customer, defining a delivery zone — is a string equality join or a bounded neighborhood lookup (`k-ring`), both native to columnar warehouses. The computational profile of spatial analytics collapses to that of ordinary SQL.

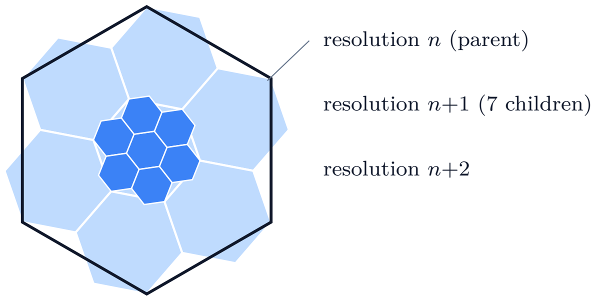

Figure 1: Aperture-7 hierarchical nesting of the H3 grid: each parent cell at resolution n subdivides into approximately seven children at resolution n+1; truncating an index string coarsens the view.

2.2 Tooling Ubiquity

The pragmatic case for H3 is that the entire enterprise toolchain now speaks it natively. Cloud warehouses ship H3 functions in SQL [10, 11]; desktop GIS added first-class support when ArcGIS Pro 3.1's *Generate Tessellation* tool gained H3 grid types, with Business Analyst workflows built on the same cells [3]; and browser renderers (Kepler.gl, deck.gl, Felt) draw H3 layers from a plain column of indexes. An organization can adopt the spine without licensing a single new component.

2.3 Retail-Geography Lineage

The analytical layer we build on the spine has deep roots. Reilly's law of retail gravitation [6] and Zipf's interaction hypothesis [8] established that commercial attraction decays with distance and scales with mass; Huff's probabilistic trading areas [7] gave the canonical store-choice formulation. Our revenue-capture model (Section 5) is a Huff-style decay kernel evaluated on hexagon centroids, and our whitespace classification is the modern, warehouse-executable form of trading-area gap analysis. The MAUP critique of Section 3 follows Openshaw [5].

3 The Case Against Administrative Geography

Administrative polygons exist to govern, not to measure markets. Used as analytical units they introduce three distinct, compounding distortions.

3.1 The Modifiable Areal Unit Problem

A *desa* in dense urban Jakarta can cover under 1 km^2 and hold 30,000 residents; a *desa* in Kalimantan can span hundreds of square kilometers with fewer than 1,000. Any choropleth of revenue, population, or penetration built on such units is dominated visually and statistically by polygon *size* rather than by the phenomenon being mapped: vast sparse districts swamp the map while the urban cores where the business is actually won or lost vanish into slivers. Figure 2 shows the effect on our study corridor with the underlying data held constant — district-level aggregation (panel a) manufactures large homogeneous plateaus and erases the demand gradients that the uniform resolution-8 grid (panel b) preserves. The distortion is not a visualization nuisance; it propagates into every downstream ratio. Standardizing the archipelago onto $\sim 0.737\,\mathrm{km}^2$ hexagons makes "revenue per cell" an apples-to-apples quantity from Jakarta to Kalimantan.

3.2 Cross-Border Catchment Blindness

Consumers do not respect political borders. A store on the western edge of DKI Jakarta draws naturally from Tangerang in Banten province; a distribution hub near a provincial boundary serves both sides. Grouping metrics by *kota* erects artificial walls through these organic catchments: the store's performance is judged against the population of the wrong polygon, and the demand it actually serves is credited to a neighbor. On the spine, a catchment is simply the set of cells within k rings of a location — a neighborhood query that crosses provincial borders as indifferently as consumers do (Section 5, Figure 4).

3.3 *Pemekaran* Instability

Indonesian administrative geography is a moving target: provinces, regencies, and villages split (*pemekaran*) with regularity. Each split breaks the join key of every historical dataset aggregated to the old boundaries — year-over-year comparisons must either be abandoned or rebuilt through error-prone crosswalk tables. H3 cells, by contrast, are mathematically fixed: the cell containing a given coordinate at a given resolution is the same today, a decade ago, and a decade hence. A spine keyed to H3 makes the organization's spatial history immune to political redistricting by construction.

4 The H3 Spine Architecture

The spine is deliberately boring technology: one dimension table and a discipline about when indexing happens. Figure 3 shows the pipeline; it stays entirely inside the existing warehouse.

Table 1: Notation.

Symbol	Meaning
H, S	set of spine cells / stores
P_h	population of cell h
a_s	attractiveness (gross pull) of store s
$d(h, s)$	hex-grid distance, cell h to store cell
τ	catchment decay length (cells)
κ	spend conversion rate (IDR/person)
R_h	monthly revenue captured from cell h
r_h	revenue per capita, R_h/P_h
P^*, r^*	population / rev-per-capita thresholds
$\mathcal{K}_k(s)$	k-ring catchment of store s
$G(x)$	population reached at area share x
W_h	composite demand mass of cell h

Index at ingestion, never at query time. Coordinates from POS systems, ERP store masters, and field devices are converted to a resolution-8 H3 string the moment they land. Geocoding debt is paid once; every analyst thereafter inherits a clean join key. External layers — open population data already distributed on H3 cells [9], competitor registries, demographics — are transformed to the same resolution on arrival.

One spine, many resolutions. The master dimension holds every cell covering the operating territory. Because aperture-7 parentage is a function of the index, coarser rollups are derived columns, not separate pipelines. The spine also carries slowly changing cell attributes (urban/rural class, region ownership, serviceability flags) so that business rules join at the same key as facts.

Metrics as materialized views. The analytical layer of Section 5 compiles to ordinary SQL over the spine — group-bys, window percentiles, and bounded `k-ring` expansions — maintained as scheduled materialized views. BI and GIS tools connect to these views and render *pre-aggregated* cell metrics, which keeps dashboards fast, load predictable, and governance centralized: there is exactly one definition of "whitespace" in the company, and it lives in the warehouse, not in a workbook.

5 Analytical Method

We now define the metrics layer. All quantities are per-cell; h ranges over spine cells and s over stores. Table 1 collects the notation.

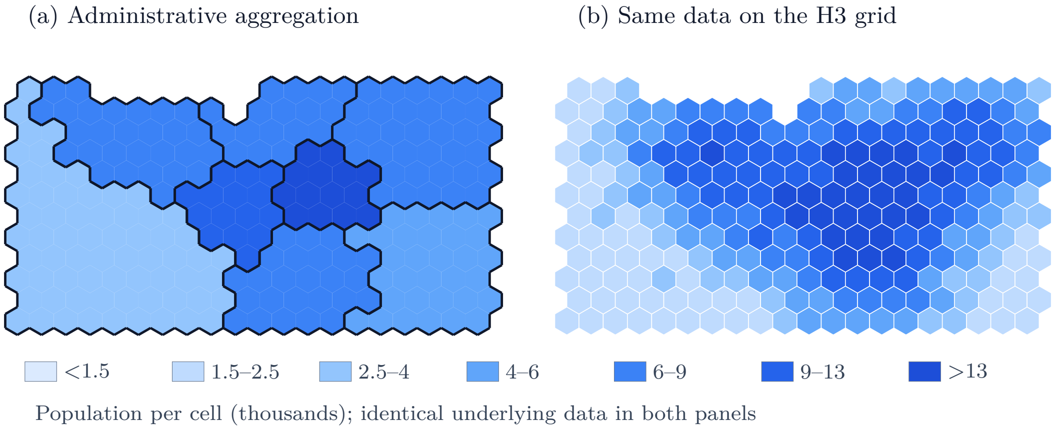

Figure 2: The modifiable areal unit problem on the study corridor. Both panels color the identical synthetic population surface; (a) aggregates it to irregular administrative districts, (b) renders it on the uniform H3 resolution-8 grid.

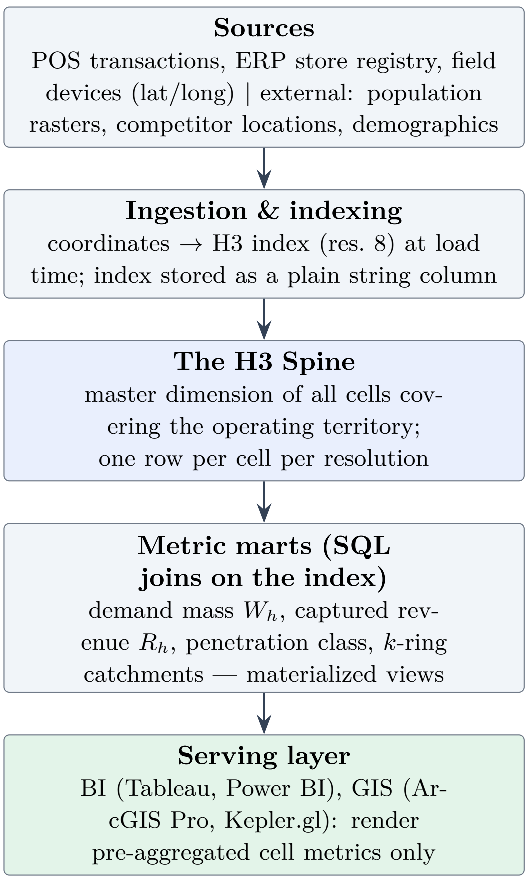

Figure 3: The H3 Spine pipeline. Indexing happens once at ingestion; every downstream operation is a join on the H3 string, and visualization tools render pre-aggregated metrics rather than raw geometry.

5.1 Demand Surface

The base layer is a population (or, more generally, demand) surface on the spine. Where richer signals exist, a composite mass generalizes raw population:

$$W_h \;=\; \lambda_P P_h + \lambda_O O_h + \lambda_S S_h, \tag{1}$$

with O_h historical order volume, S_h serviceable-outlet density, and λ's business-set calibration weights — the same composite that feeds the routing layer in [12]. The case study uses the pure population surface ($\lambda_O = \lambda_S = 0$), which is the honest starting point for an organization bootstrapping the spine from open data [9].

5.2 Revenue Capture with Distance Decay

Each store projects a Huff-style [7] Gaussian decay kernel onto the surrounding cells; the revenue captured from cell h is

$$R_h \;=\; \kappa\, P_h \sum_{s\in S} a_s \exp\!\left(-\frac{d(h,s)^2}{2\tau^2}\right), \tag{2}$$

truncated to zero below a floor ε (a cell whose entire predicted capture is negligible is treated as unserved). The decay length τ is the single most consequential calibration: it encodes how far consumers travel to a store, and should be fit per store format from loyalty or delivery-address data. Note that $d(h,s)$ is the *hex-grid* distance — a constant-time index computation on the spine, not a road query; where road realism matters the kernel can be re-evaluated on travel-time isochrones without changing anything downstream.

5.3 Penetration Classification

Cells are classified by crossing the demand surface with captured revenue. With P^* a population percentile threshold (the case study uses the 60th) and r^* a per-capita revenue threshold (40th percentile of served cells), rules evaluated top to bottom:

$$\text{class}(h) = \begin{cases} \text{low priority}, & P_h < P^*, \\ \text{whitespace}, & R_h = 0, \\ \text{under-penetrated}, & r_h < r^*, \\ \text{penetrated}, & \text{otherwise.} \end{cases} \tag{3}$$

The taxonomy is deliberately coarse — four classes an executive can act on. *Whitespace* (people, no revenue) is the expansion shortlist; *under-penetrated* (people, weak revenue) is the marketing and assortment shortlist; the thresholds are percentile-based so the classification survives recalibration of κ and a_s unchanged. The monetary size of the whitespace follows directly: at the served-cell average revenue per capita $\bar{r}$, the opportunity is $\bar{r}\sum_{h\in\text{white}} P_h$.

5.4 k-Ring Catchments and Cannibalization

The k-ring $\mathcal{K}_k(s) = \{h : d(h,s) \leq k\}$ is the spine's native catchment primitive. For two stores, the shared cells $\mathcal{K}_k(s)\cap\mathcal{K}_k(s')$ measure catchment overlap; the population-weighted overlap share

$$\chi(s,s') = \frac{\sum_{h\in\mathcal{K}_k(s)\cap\mathcal{K}_k(s')} P_h}{\sum_{h\in\mathcal{K}_k(s)} P_h} \tag{4}$$

is a direct cannibalization index, computable before a candidate site is committed (Figure 4). The same primitive defines fulfillment zones for distribution centers and dispatch clusters for field-service teams.

5.5 Targeting Gain Curves

Finally, the spine makes targeting efficiency measurable. Rank cells by descending P_h and let $G(x)$ be the share of total population contained in the top x share of land area; the administrative counterpart ranks whole districts by mean density and accumulates entire polygons. The gap between the two curves (Figure 9) is the price of being forced to buy geography in district-sized blocks — it bounds the achievable precision of geofenced media, sampling campaigns, and coverage commitments under each geography.

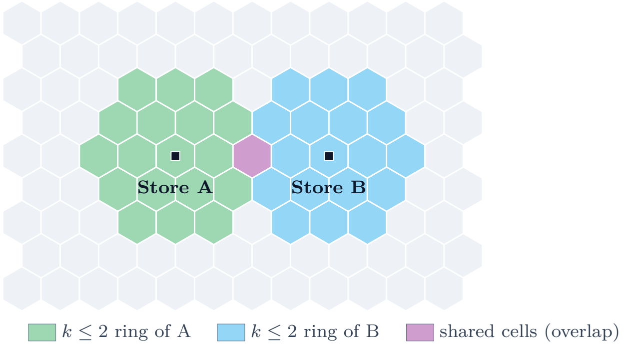

Figure 4: k-ring catchment overlap between two stores four cells apart: cells within $k \leq 2$ of each store, with the shared cell marking the cannibalization zone.

6 Case Study: The Western Jakarta–Tangerang Corridor

We exercise the full spine workflow on a deterministic synthetic corridor styled on the western Jakarta–Tangerang conurbation: 232 land cells at resolution 8 ($\sim$171 km^2) between a masked Java Sea coastline and an inland fringe, holding 1.61 million synthetic residents. The surface composites five density centers — a CBD core, western ("Tangerang-side"), eastern, and southern sub-centers, and an infill corridor — over a noisy urban baseline. Twelve stores (S01–S12) cluster in the CBD, the west, and the south; *the eastern sub-center has none*, a deliberate gap the analysis should rediscover. All data are generated by a seeded script (`generate_figures.py`), so every number and figure in this section is exactly reproducible; the workflow, not the synthetic magnitudes, is the claim being demonstrated.

6.1 Layer 1: Population

Figure 5 shows the demand surface with the store network overlaid. Even at a glance the resolution-8 grid conveys what district aggregation cannot (compare Figure 2a): the CBD gradient, the three secondary peaks, and the store network's western bias are all legible because every cell is the same size.

6.2 Layer 2: Revenue, Joined on the Spine

Applying the capture model of Eq. (2) with decay length $\tau = 1.55$ cells ($\sim$1.5 km) yields IDR 507.8 bn of monthly revenue attributed across cells — Figure 6 renders it as bubbles over the population fill. The join that produces this figure is one line of SQL on the spine: revenue facts and population both keyed by the same H3 string. The overlay already telegraphs the finding: the eastern population peak carries essentially no revenue mass, and its cells are outlined as whitespace.

6.3 Layer 3: Classification

Applying Eq. (3) with the 60th-percentile population threshold ($P^* = 7{,}076$ per cell) and the 40th-percentile per-capita revenue threshold ($r^* =$ IDR 136,472/person/month) partitions the corridor into 77 penetrated, 13 under-penetrated, 3 whitespace, and 139 low-priority cells (Figure 7). The scatter view (Figure 8) shows the same partition in metric space: whitespace cells sit far right on the population axis at exactly zero captured revenue — populous, unserved, and invisible to any analysis keyed on served territory.

6.4 Sizing the Opportunity

The three whitespace cells hold 23,210 residents — only 1.4% of corridor population, but concentrated in the eastern sub-center at walkable density. At the served-cell average of IDR 361,662 per person per month, closing them is worth ∼IDR 8.4 bn of monthly revenue, before counting the 13 under-penetrated cells (136,164 residents currently monetizing below r^*). This is the strategic deliverable of the spine: not a map, but a ranked, priced shortlist of cells — each ∼0.74 km^2, small enough to visit, audit, and act on.

7 Targeting Efficiency and Field Applications

7.1 The Price of District-Sized Blocks

The gain curves of Figure 9 quantify what administrative targeting costs. Ranking resolution-8 cells by density, 50% of corridor population is reached within 25.4% of the land area (43 km^2); accumulating whole districts requires 38.4% (66 km^2) for the same reach. At the 75% reach level the gap persists: 47.0% of area (80 km^2) for hexagons versus 57.8% (99 km^2) for districts (Figure 10). Every campaign whose cost scales with area — geofenced digital media, leaflet drops, field sampling, canvassing coverage — runs ∼1.2–1.5× leaner on the spine, and the advantage widens with the size disparity of the underlying districts, which on the real archipelago is far larger than in this stylized corridor.

7.2 Retail Footprint

The classification map is the expansion agenda: whitespace cells are site-selection candidates, scored by P_h and validated in the field. Before committing a site, Eq. (4) prices its interaction with the existing network — in the stylized configuration of Figure 4, two stores four cells (∼3.7 km) apart at $k \leq 2$ share a single cell, roughly 5% of each catchment, a comfortable spacing; candidate sites closer to existing stores see χ rise steeply. The same join against an indexed competitor registry yields competitor-density covariates for revenue modeling.

7.3 Network Logistics

Distribution-center fulfillment zones are k-ring (or isochrone-refined) cell sets with known population, order mass, and revenue — so an SLA commitment is a measurable statement about a specific set of hexagons rather than an aspiration about a *kecamatan*. Daily delivery manifests group naturally by cell for route assignment, and demand staging by SKU follows the per-cell order history. The handoff to fleet optimization is direct: the spine's demand masses W_h are exactly the gravitational masses consumed by the routing engine of [12].

7.4 Dynamic Resource Allocation

Because spine metrics refresh as materialized views, the same machinery serves operational tempo: field-service and hypercare technicians assigned to cell clusters during go-lives, and supply–demand imbalances (transport availability versus order volume per cell) triggering surge responses in near real time. The unit of allocation is always the same auditable object — an H3 cell — from strategy deck to dispatch board.

8 Implementation Roadmap

The spine earns trust incrementally; each phase ships a usable artifact.

Phase 1 — Proof of concept (one region, one KPI). Build the spine for a single high-density region (Greater Jakarta), join internal store revenue against open population data [9] at resolution 8, and publish one governed metric — revenue per capita per cell — with the classification map of Eq. (3). Everything in Section 6 is achievable in this phase with no new infrastructure; the deliverable is the whitespace shortlist and the stakeholder conversation it starts.

Phase 2 — Pipeline automation. Move indexing to ingestion: nightly batch transforms convert raw POS/ERP coordinates into H3 strings inside the warehouse, the spine dimension and metric

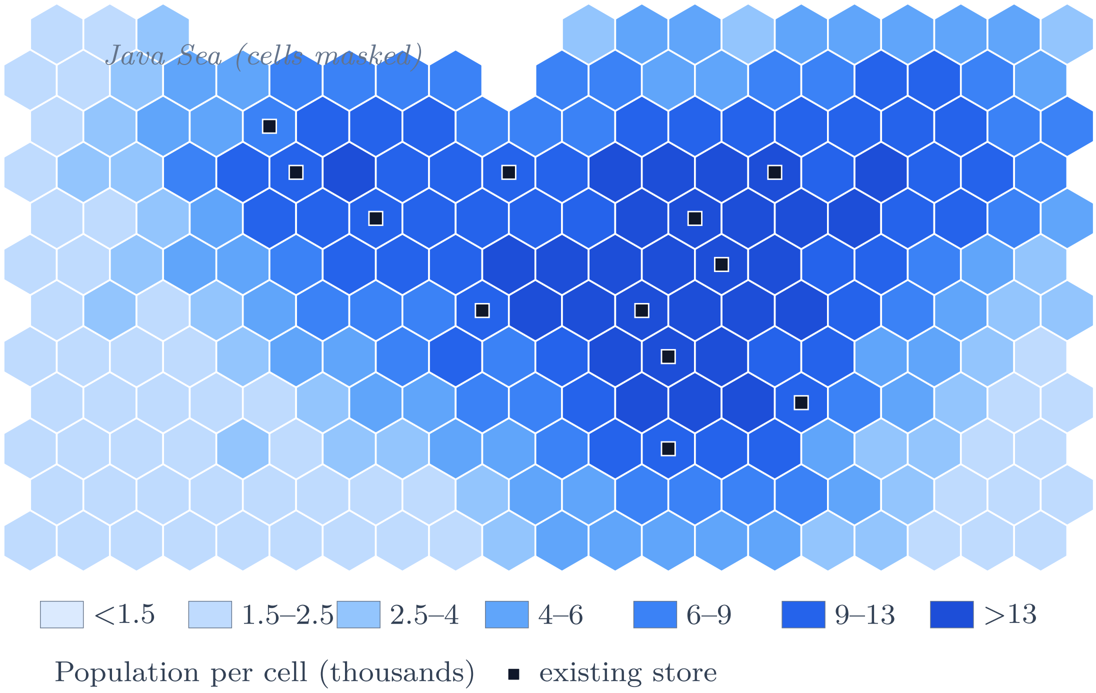

Figure 5: Synthetic population surface of the study corridor on H3 resolution-8 cells ($\sim$0.74 km^2 each), with the existing store network overlaid.

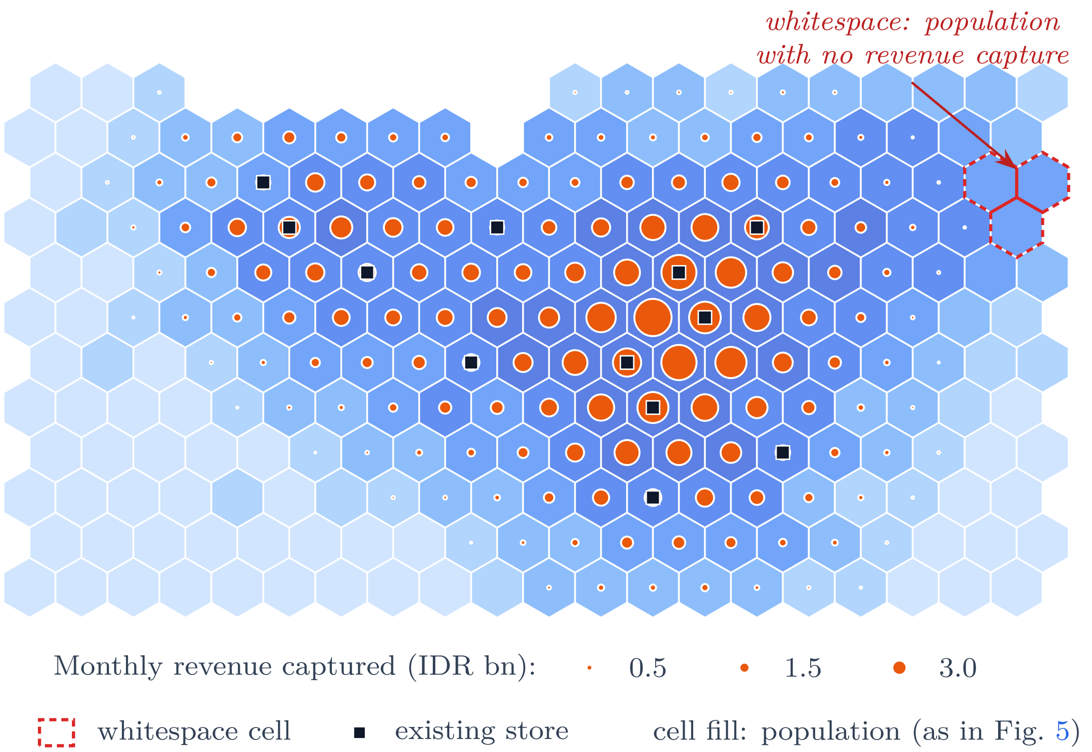

Figure 6: Joining the two datasets on the spine: population fill, captured monthly revenue as bubbles, and whitespace cells outlined.

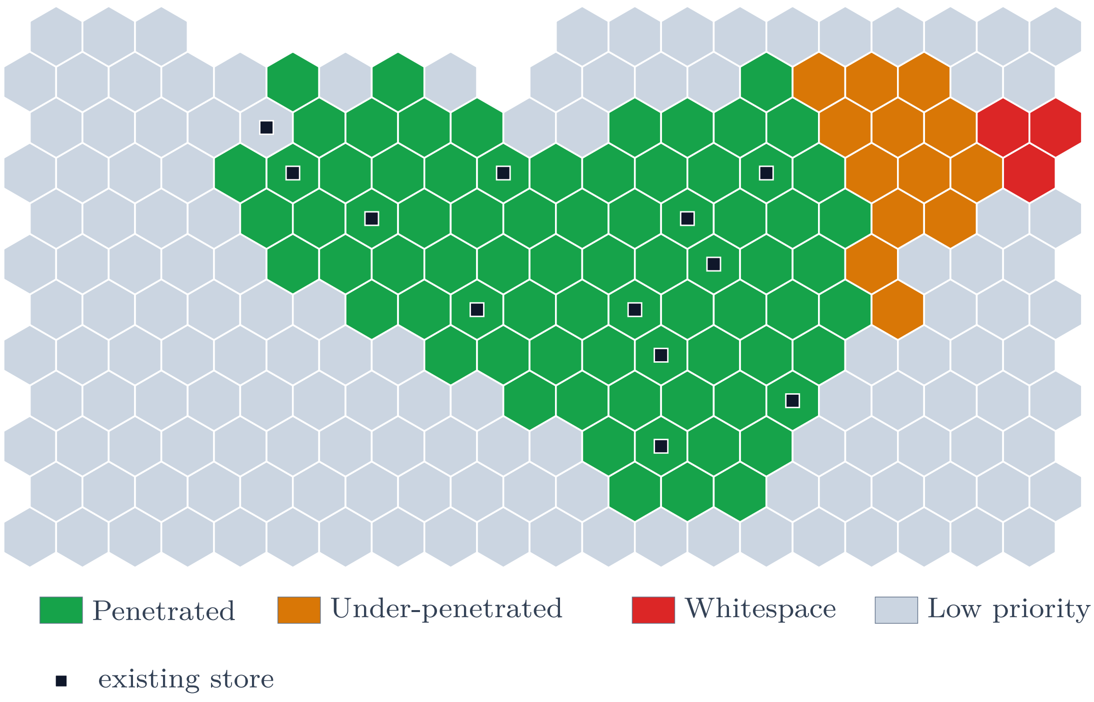

Figure 7: Cell-level penetration classification derived from the joined population and revenue layers.

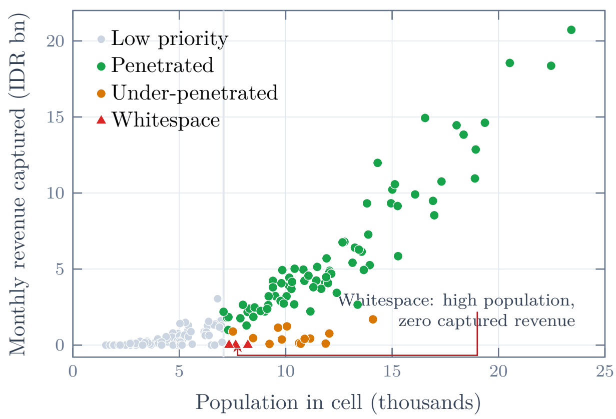

Figure 8: Population versus captured revenue per cell, colored by classification; whitespace cells sit on the zero-revenue axis at high population.

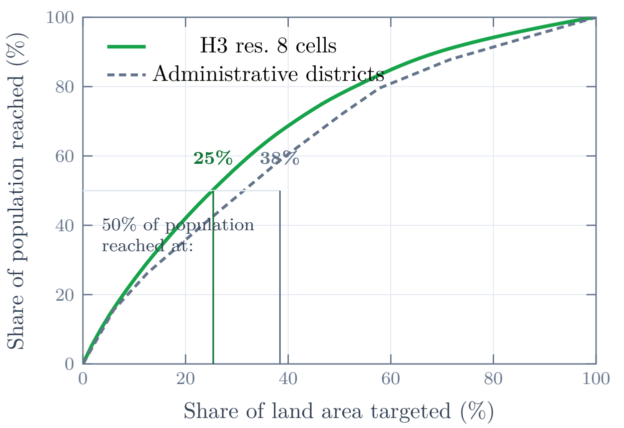

Figure 9: Population-reach gain curves: share of corridor population reached as a function of land area targeted, cell-by-cell on H3 versus whole administrative districts.

marts become scheduled materialized views, and the classification thresholds (P^*, r^*) move from notebook constants to versioned configuration. Data-quality gates (geocode validity, coordinate-swap detection, cell-coverage drift) are attached to the load, because from this phase on the spine is load-bearing.

Phase 3 — National scale and advanced primitives. Extend the spine across the operating territory, activate k-ring analytics (cannibalization screens, DC fulfillment zones), calibrate the decay length τ and store attractiveness a_s per format from delivery and loyalty data, and connect downstream consumers — media geofencing exports, dispatch clustering, and the routing engine of [12]. Regional divergence matters at this stage: dense Java corridors and sparse eastern territories warrant different working resolutions, selected per region exactly as the companion framework selects routing parameters.

9 Limitations and Future Work

Synthetic calibration. The corridor is a stylized artifact: magnitudes (ARPU, opportunity sizing) illustrate the workflow, not the market. The immediate next step is re-running the identical SQL

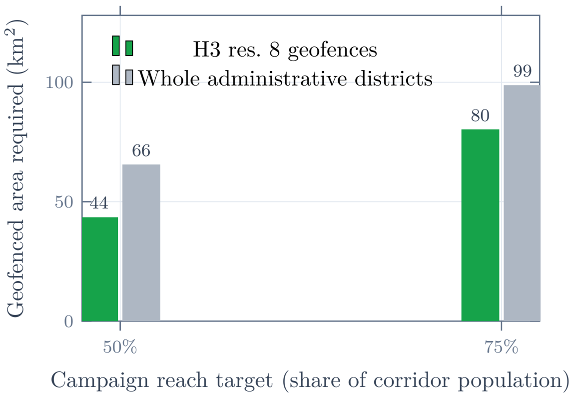

Figure 10: Geofenced area required to reach 50% and 75% of corridor population under hexagon-level versus district-level targeting.

against production POS data and Kontur population — the figures regenerate unchanged because they consume only the spine schema.

Grid-distance decay. Eq. (2) decays with hexgrid distance, which ignores rivers, toll barriers, and one-way systems. Isochrone-refined kernels (travel-time bands from the routing stack) are a drop-in replacement where site decisions are close.

Ecological inference. Cell-level correlation is not customer-level behavior; resolution-8 aggregates ∼thousands of residents, and inferences about individuals from cell metrics commit the ecological fallacy. The spine bounds, it does not replace, customer-level analytics.

Boundary effects at coarse rollups. Aperture-7 parentage is approximate at cell edges; metrics rolled up several resolutions can shift small mass between parents. For executive rollups this is immaterial; for contractual SLA geometry, compute at the working resolution.

Threshold sensitivity. The four-class taxonomy inherits the percentile choices (P^*, r^*); a sensitivity sweep belongs in the Phase-2 automation so that the whitespace list ships with its stability band.

10 Conclusion

Strategic market analysis fails quietly when its spatial unit is wrong: district choropleths that hide urban demand, catchments amputated at provincial borders, and histories severed by *pemekaran* are not data problems but geometry problems. The H3 Spine fixes the geometry once, at ingestion, and lets everything downstream — population joins, revenue capture, penetration classification, cannibalization screens, targeting gain curves — run as ordinary SQL on one shared key. On a reproducible Jakarta–Tangerang corridor the approach rediscovered a deliberately planted market gap, priced it at ∼IDR 8.4 bn per month, and cut the area cost of reaching half the population by a third relative to district targeting. The same cells then feed routing, dispatch, and geofencing without translation. Hexagons are not the insight; the discipline of one fixed, uniform, hierarchical spatial key under every layer of the business is — and the corridor study shows it can be stood up with open data and a weekend of SQL.

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